

Part B-1

1. Excellence

1. Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

Introduction.

Unlike animals, whose body plan is essentially fixed during embryogenesis, plants build most of their body after germination. This post-embryonic mode of development is the source of their remarkable plasticity, and it depends on the sustained activity of meristems, a specific tissue containing stem cells that give rise to all plant organs.

The shoot apical meristem (SAM) originates all aerial organs leading to the initiation of leaves, stems and flowers; during vegetative development, its activity sets the pace for leaf initiation rate, known as the **plastochron**. Temperature is a central regulator of this rate: within a biologic range, warmer temperature accelerates leaf production, and this acceleration is so regular that the rate of leaf initiation is linearly related to the **thermal time**¹: the amount of degrees the plant experience over a period of time. This relationship is conserved from Arabidopsis to crops and is a major determinant of development and yield. While temperature signaling has been dissected in detail in the context of hypocotyl elongation, its action within the SAM, where the organs that shape the plant are born, **remains almost completely unexplored**. This project aims to decipher how the SAM perceives and transduce temperature cues to set the plastochron and proposes that a missing piece might be the redox and nitric oxide (NO) signaling.

State of the art and objectives.

The SAM is organized into functional zones. In the central zone, a WUSCHEL (WUS)-CLAVATA3 (CLV3) feedback loop maintains stem cell identity; in the peripheral zone, faster-dividing cells differentiate into leaf primordia at sites specified by PIN1-dependent auxin maxima, with cytokinin promoting leaf proliferation. Recent 3D imaging showed that temperature acts directly within the SAM, altering meristem size and WUS/CLV3 expression patterns (Wenzl and Lohmann, 2023). However, **how these changes drive leaf initiation, whether temperature signals through auxin or cytokinin pathways, and how temperature perception is first transduced into SAM activity remain unknown**.

Temperature-sensing has been described in the hypocotyl: phytochrome B (phyB) acts as a thermosensor inactivated at warm temperature; ELF3, a transcriptional regulator, also gets inactivated under warm temperature; PIF7 whose transcription increases through temperature-dependent mRNA translation; and warm temperature drives starch breakdown and sucrose production. Downstream, signaling converges on PIF4 and the E3 ligase COP1. Whether these mechanisms also operate in the SAM to control leaf initiation is unknown.

In parallel, it is known that redox species constitutes key signals within the SAM. The meristem is also redox-compartmentalised: superoxide accumulates in the central zone, sustaining WUS and stemness, whereas hydrogen peroxide and NO are enriched in the differentiating peripheral zone, and disrupting this gradient reduces SAM size and delays early leaf development (Zeng et al., 2017; Zeng et al., 2023). The peripheral zone is precisely where primordia are initiated, so the NO-rich domain coincides with the site where the plastochron is literally set. NO promotes peripheral-zone fate by repressing WUS, in part by limiting the stem-cell regulator AGO4 to the centre through S-nitrosylation, with WUS interacting with AGO4 in a NO-dependent manner (Zeng et al., 2023). The meristem redox state

is further shaped by its microenvironment: the stem-cell niche is hypoxic, and low oxygen is itself a developmental signal transduced

Preliminary results.

Hypothesis and objectives. My hypothesis is that temperature is transduced in the SAM through a redox/NO node that remodels the peripheral-zone redox landscape to set the plastochron. **The general aim is to understand the early mechanisms linking temperature perception to plastochron regulation in the SAM.**

WP1: Characterization of temperature on early development in the meristem. I aim to describe the effect of temperature on SAM function and leaf initiation with high spatial and temporal resolution (plastochron on thermal-time basis, SAM size, WUS/CLV3 expression and distribution).

WP2: determine the role of phyB and of redox/NO signalling in the temperature control of leaf initiation, addressing the general redox state of the meristem (2.1) and nitric oxide specifically (2.2).

WP3: characterise the transcriptomic response of the SAM to warm temperature at cellular resolution, testing whether known meristem regulators participate in temperature and thermal-time sensing.

Originality and ambition. The project moves temperature signalling out of the hypocotyl and into the meristem; it proposes a specific, testable transducer (redox/NO) rather than a generic unknown, grounded in preliminary data showing the canonical module is dispensable; and it connects thermomorphogenesis, which has the sensors, with the redox biology of the stem-cell niche, which has the positional signals. Answering how environmental information is converted into the timing of organ formation is of broad significance for developmental biology and directly relevant to crop resilience under a warming climate, where leaf-initiation rate shapes canopy establishment and yield.

2. *Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).*

Overall approach. The methodology pairs two logics rarely combined in one laboratory: the quantitative, imaging-based developmental biology of the SAM, and the biochemistry of NO and redox signalling. A single output, the rate and pattern of organ initiation expressed on a thermal-time basis, is read through three complementary lenses (genetic/phenotypic, in vivo imaging, biochemical), so that no central claim rests on a single technique.

By objective. For WP1, I quantify plastochron and phyllotaxis against thermal time across wild type and mutants (vegetative phase only, controlling for flowering time), computing thermal time from logged chamber temperatures and comparing per-genotype linear models statistically; live confocal imaging of stem-cell (pCLV3, pWUS) and auxin (DR5, PIN1) reporters under temperature controlled at the sample with the CherryTemp system available in the host laboratory, used both for stable set-points and for rapid temperature shifts during time-lapse. For WP2, I map redox species in the living apex: the O₂^{•−}/H₂O₂ balance (DHE/NBT and H₂DCFDA/DAB) and NO (DAF-FM DA), across temperatures and in phyB, asking whether temperature reconfigures the central-versus-peripheral compartmentalisation; I assess the hypoxic niche and the N-degron/ERF-VII route as a function of temperature; and I test function by genetic (gsnor1/hot5, nia1nia2, noa1) and pharmacological perturbation. Preliminary pharmacological experiments manipulating NO with the donor GSNO and the scavenger cPTIO are underway, measured on a thermal-time basis and directly comparable with the phyB phenotype; glutathione is included as a control for the general thiol-redox context rather than as an NO-specific treatment. The decisive experiment asks whether NO perturbation reproduces the loss of thermal-time compensation seen in phyB. The membrane-level phyB–RBOHD–FERONIA

precedent (Mohanty et al., 2026) is tested in the SAM context. For WP3, single-cell RNA-seq under a factorial temperature $\times$ thermal-time design (below) yields the temperature- and redox-responsive networks of the SAM, analysed against known meristem gene sets. Methods across WP2 also include knock-out mutants crossed with DR5, PIN1, WUS and CLV3 reporters, histochemical staining and confocal quantification, characterising the temperature response in the core.

scRNA-seq design. Comparing two temperatures at equal chronological age confounds temperature with developmental stage (warm develops faster); comparing at equal thermal time controls development but not chronological age. I therefore use a factorial design, two temperatures (control vs warm) $\times$ two equivalent thermal-time points, rather than a gradient, which for single-cell would dilute replication and power. This separates the effect of temperature at matched development from the effect of developmental progression, and provides a direct test of compensation: a gene genuinely compensated on a thermal-time basis should not change between temperatures once thermal time is matched. Predictions from Zeng et al. (2017, 2023) anchor quality control and the analysis panel (WUS/CLV3 targets, AGO4 and RdDM components, zonal ROS-metabolising enzymes, NO-biosynthesis genes NIA1/NIA2/NOA1, auxin readouts).

Interdisciplinarity. The project sits at the interface of developmental biology, plant physiology and redox biochemistry; its feasibility rests on combining expertise that does not currently coexist in one place, the host's developmental and photobiological framework and quantitative imaging, and my redox/NO biochemistry and in vivo detection.

Open science. Data are managed under a FAIR-compliant Data Management Plan prepared in the first months and updated thereafter; datasets underlying publications (imaging, quantitative phenotyping, single-cell data) are deposited in recognised repositories; publications are made immediately open access via Libra, the UniNE institutional repository, and disciplinary preprint servers at submission; protocols and analysis code, including the thermal-time image-analysis and statistical pipelines, are shared openly. Where possible, confirmatory analyses are pre-registered.

Sex and gender dimension. This project is performed in the plant *Arabidopsis thaliana*, so a sex and gender dimension does not apply to its research methodology; however, gender dimension is considered.

3. Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

What I gain from the host. My doctoral training in Argentina was in NO and redox signalling in plant physiology, at the level of the whole plant and of biochemical readouts. This fellowship provides what I do not yet have: the capacity to ask developmental questions at cellular resolution and in real time. This research group works at the interface of light and temperature signalling and shoot development, making it the environment where I can convert my signalling expertise into developmental biology. **My training plan centres on three competences:** quantitative live-imaging of the meristem (reporter lines, temperature-controlled imaging with CherryTemp, 3D image-analysis pipelines); the genetics and conceptual framework of thermomorphogenesis, a field to which my supervisor is a foundational contributor; and single-cell transcriptomics with its computational analysis. Formal training is supported by the UniNE Graduate Campus (transferable-skills workshops and career development, part of the swissuniversities 2025–2028 programme), the doctoral programmes of UniNE and the CUSO, and [EMBO/other courses to specify].

What I bring to the host. The host group has no in vivo redox capability. I bring detection of reactive redox species in living tissue (DAF-FM DA for NO, H2DCFDA and DHE for ROS), the genetics of NO homeostasis, pharmacological manipulation of NO with donors and scavengers, and the biochemistry of cysteine-based post-translational modifications including biotin-switch workflows. The group's interest in redox as an integrator of environmental and endogenous cues in the SAM is explicit and current, but the technical and conceptual capacity to pursue it is not yet established in

Neuchâtel; my arrival makes that line executable and installs a capability that outlives the fellowship. This is a genuine two-way transfer.

Quality of supervision. Prof Martina Legris leads a research group at the Institute de Biologie, University of Neuchâtel. She specialises in plant photomorphogenesis and thermomorphogenesis, with [>20] publications on phytochrome and photoreceptor signalling, and her current interest is the regulation of meristematic activity by environmental cues such as shade and warm temperature. Her contribution to the field is foundational: she is the first author of the work establishing phyB as a thermosensor integrating light and temperature in Arabidopsis (Legris et al., Science, 2016), and she has developed the study of light and temperature control of leaf and meristem morphogenesis (Legris et al., Nat Commun, 2019; Legris, New Phytologist, 2023). She trained in the Casal and Fankhauser laboratories and is herself a former MSCA and EMBO fellow, so she knows the fellowship from the inside. My project is a direct extension of a question her own work opened, ensuring the supervision is substantive. [Completar: composición actual del grupo; estudiantes/postdocs supervisados; financiamiento activo (buscar en data.snf.ch); arreglo de reuniones semanales; plan de mentoreo de un estudiante de máster en el año 2.]

Secondment.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

I obtained my PhD in Exact Sciences (Biological Sciences and Biotechnology) with special distinction from the National University of La Plata in 2025, at the Institute of Plant Physiology (INFIVE, UNLP-CONICET), on the role of NO in acclimation to phosphorus restriction. I established that nitrate reductase mediates NO generation and that this NO is required for acclimation responses, published as first author (Luquet et al., Plant Science, 2025), and contributed to five further papers including reviews that helped define the agenda linking NO signalling to nutrient use efficiency. My work received the Best Poster Prize at the XXXV Meeting of the Argentine Society of Plant Physiology (2025) and I was awarded a competitive travel grant to present the flash talk “At the apex of thermomorphogenesis” at the 2026 thermomorphogenesis meeting in Dundee, which reflects the reception of this new line of work by the very community this project addresses.

I bring to the lab first, technical skills built by necessity: I established and applied in vivo redox imaging, enzymatic and HPLC determinations, biochemical analysis and hydroponic physiology, in a setting where techniques are more often built than bought. Second, early independence: I supervised an undergraduate project student for three years, taught as a teaching assistant, and sought out my own collaborations with diverse groups, including collaborations with Lamattina's group in Mar del Plata (2021) which resulted in a research paper published in 2025 (CITA) and a research stay in the Chromosome Dynamics group at IPS2, Université Paris-Saclay (2025), whose chromatin-biology background is directly relevant to the AGO4/DNA-methylation dimension of this project. Third, sustained science outreach, which I will continue in Neuchâtel.

My doctorate gave me a clear identity as a redox biologist, and this fellowship is the moment to put that identity to work on questions I could not ask alone. Moving into developmental biology is a deliberate choice: redox signalling is already recognised as a regulator of meristem activity, but the tools to follow it in vivo and to connect it to temperature are not established there, and that is what I bring. Throughout my training in Argentina and in France I learned that science moves through people, and that open, collaborative practice is what makes knowledge grow. That is why I share materials and protocols openly and invest time in outreach, and why I intend to publish this project's results in open access and to deposit its data and protocols in public repositories. My long-term objective is to lead an independent group on how environmental signals are transduced by redox chemistry into developmental decisions.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

My long-term objective is to lead an independent research group working at the interface of redox biology and plant development. This fellowship is the decisive step towards that goal, because it adds to my established expertise in nitric oxide and redox physiology the three competences I currently lack and that this research programme requires: quantitative live-imaging of development at cellular resolution, the genetics and conceptual framework of thermomorphogenesis, and single-cell transcriptomics together with its computational analysis. Acquiring these in a group that works at the interface of light, temperature and shoot development positions me to lead a distinctive line of research that few others can, precisely because it joins two communities that rarely meet.

My skills development is organised around a Career Development Plan, agreed with my supervisor during the first months of the fellowship and reviewed at regular intervals, which sets objectives for research training, transferable skills and career progression. Scientific training follows the work packages themselves: WP1 and WP2 build my competence in reporter-based confocal imaging, temperature-controlled live-imaging and image quantification, while WP3 introduces me to single-cell transcriptomics and the bioinformatic analysis of complex datasets, an increasingly indispensable skill in plant developmental biology. The secondment to [the Weits group, Utrecht University] extends this further by giving me direct training in hypoxia and oxygen-sensing biology, a dimension of meristem physiology I could not access otherwise.

Transferable skills are supported by the host institution's provision. The UniNE Graduate Campus, established as part of the swissuniversities programme for promoting academic and scientific talent (2025–2028), offers workshops and courses designed to develop transferable skills and support the career development of postdoctoral researchers, and I will also have access, subject to availability, to courses offered by the doctoral programmes of UniNE and by the CUSO. I will use this provision to strengthen the competences an independent researcher needs beyond the bench: scientific writing and grant preparation, project and data management, research integrity, and leadership and supervision. I will further participate in the Réseau romand de mentoring pour femmes and the REGARD workshop programme, which support the progression of women towards independent academic positions.

Supervision experience is a deliberate part of the plan. During the second year I will propose and supervise a master student on a defined part of the project, with the support of my supervisor, which will give me first-hand experience of mentoring and of designing a project at the right scale for a student. Together with the teaching experience I already have from Argentina, this prepares me for the training responsibilities of a group leader.

Finally, the fellowship substantially widens my network. Beyond the host group and the secondment, Switzerland's position and the host's collaborations give me access to the European plant science community, and I will present my results at international conferences [specify: ICAR, EPSO, thermomorphogenesis or plant development meetings] and at institutional seminars. Combined with my international trajectory across Argentina, France

and Switzerland, and with a portfolio that will by then span redox biochemistry, developmental imaging and single-cell genomics, this makes me competitive for independent-investigator schemes such as SNSF Ambizione or Starting Grants and the ERC Starting Grant, whether I continue my career in Europe or return to Latin America, where these competences are scarce and where I would be able to establish them.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination. The primary route for disseminating results is peer-reviewed publication. I anticipate at least two publications: a first, based on the characterisation of the temperature response of the SAM (WP1) and the redox and NO perturbation experiments (WP2), and a second built on the single-cell dataset (WP3). Because the project sits at the intersection of two communities, I will target journals read by both plant developmental biologists and the environmental-signalling and redox fields, and I will make results available as preprints at the time of submission so that they reach these communities without delay. I will publish in open access and deposit the accepted versions in Libra, the institutional open-access repository of the University of Neuchâtel, in compliance with Horizon Europe requirements.

Results will also be disseminated through international conferences and specialised meetings [specify], through seminars at UniNE and at partner institutions, and through the secondment host. Presenting to audiences of different composition, from a specialised meristem-biology audience to a broader plant-science one, is itself part of my training in scientific communication.

Research data are a deliverable in their own right. All data underlying publications, including imaging data, quantitative phenotyping datasets and the single-cell transcriptomic dataset, will be managed under a Data Management Plan following the FAIR principles and deposited in recognised community repositories with persistent identifiers. The single-cell dataset of the SAM temperature response (WP3) is designed to be a reusable community resource: there are few single-cell datasets of the vegetative shoot apex and none, to my knowledge, addressing temperature, so its value extends well beyond the questions I ask of it here. Protocols and analysis code, including the thermal-time quantification and image-analysis pipelines, will be shared openly so that the methods are reusable.

Exploitation. The project is basic research and no intellectual property is anticipated at the outset. However, leaf initiation rate is a determinant of canopy establishment and yield, and identifying redox-based mechanisms that set it could open routes to modulate meristem activity, whether through breeding targets or through chemical modulation of redox signalling. Should any exploitable result emerge, I will assess it together with my supervisor and with the technology transfer support of the University of Neuchâtel [confirmar nombre de la oficina], and any decision on protection will be taken before publication.

Communication. Communication activities are distinct from dissemination and are directed at audiences beyond the scientific community. I have a sustained record of public engagement, including Fascination of Plants Day, university science festivals and open-science advocacy in Argentina, and I intend to continue this in Neuchâtel through the university's public engagement activities [specify]. The message I will build for general audiences connects the project to something tangible: how plants decide when to make a leaf,

and why that decision matters as temperatures rise. I will also contribute to institutional communication channels and press releases where results warrant it, and use social media to reach the wider plant-science and student community. As a woman researcher trained in Latin America and working in Europe, I also see value in making that trajectory visible to students considering research careers, and I will seek opportunities to do so.

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. The project addresses a gap that is conspicuous once stated: temperature is one of the strongest environmental regulators of plant form, and the shoot apical meristem is where plant form originates, yet the mechanisms linking the two are essentially unknown. Almost everything known about plant thermosensing comes from the hypocotyl, a tissue whose response is driven by cell expansion. By asking the same question in the meristem, where the relevant processes are stem cell homeostasis, cell division and differentiation, the project tests whether the established thermosensory framework generalises or whether the meristem uses a different logic. My preliminary data indicate the latter, which would be a substantive contribution in itself. Beyond this, the project brings redox and nitric oxide signalling into contact with thermomorphogenesis, two fields that have developed largely in parallel, and delivers a single-cell dataset of the temperature response of the shoot apex that will serve as a resource for the community. Because few groups work on environmental regulation of meristem activity, and because I will work at the interface of the developmental and the temperature-signalling communities, I expect the results to be of broad interest across plant biology.

Societal impact. The rate at which leaves are produced determines how quickly a canopy is established, and canopy establishment governs light capture, water use and ultimately yield. This relationship is conserved from *Arabidopsis* to crops such as rice and maize. Under climate change, with rising mean temperatures and more frequent warm episodes, the temperature dependence of leaf initiation becomes directly relevant to how crops perform in the field. Understanding the mechanism that sets this rate is a prerequisite for any strategy that seeks to adjust it, and contributes to the broader European objective of climate-resilient and sustainable agriculture and to the goal of food security. The project also contributes to society through training: it develops a researcher's capacity in an interdisciplinary area, and through outreach it brings plant science to non-specialist audiences.

Economic impact. No direct commercial outcome is expected within the fellowship, as this is fundamental research. In the longer term, however, knowledge of how meristem activity is set by temperature could inform breeding programmes seeking varieties with architecture optimised for a warming climate or for high-density planting, and the identification of redox-regulated components could provide targets for biotechnological modulation of meristem activity. The fellowship also contributes to the European Research Area by strengthening research capacity at the host institution, installing an *in vivo* redox capability that will remain available after the project ends, and by training a researcher whose combined skill set is scarce.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

Insert here text for your proposal

3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

Insert here text for your proposal

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Part B-2
(No overall page limit applied)

4. CV of the researcher

Insert here text for your proposal

5. Capacity of the Participating Organisation(s)

5.1 Template table: *Overview of Participating Organisations*

Organisation role	PIC	Legal Entity Short Name	Academic organisation (Y/N)	Country	Name of Supervisor
Beneficiary					
Associated partner linked to a beneficiary (if applicable)					
Associated partner for outgoing phase (mandatory for GF)					
Associated partner for secondment (if applicable)					
Associated partner for non-academic placement (if applicable)					

5.2 Template table: *Capacity of the Participating Organisations*

Maximum 1 page per Beneficiary and ½ page per Associated partner.

Choose one of:

? *Beneficiary (compulsory)*

? *Associated partner linked to a beneficiary (if applicable)*

? *Associated partner for outgoing phase (mandatory for GF only)*

? *Associated partner for secondment (if applicable)*

? *Associated partner for non-academic placement (if applicable)*

[Full name + Legal Entity Short Name + Country]	
General description	
Role and profile of supervisor	
Key research facilities, Infrastructure and Equipment	
Previous and current involvement in EU-funded research and training programmes/actions/projects	

6. Additional ethics information

Insert here text for your proposal

(NB: Only if you have additional information that could not be included in the ethics self-assessment)

7. Additional information on security screening

Insert here text for your proposal

(NB: Only if you answered yes to one of the questions in the security issues table, with the exception of “Does this activity involved HE associated and/or third countries?”)

8. Environmental considerations in light of the MSCA Green Charter

Insert here text for your proposal

9. Required for Global Fellowships only: Letter of commitment from the associated partner hosting the outgoing phase

Insert here the Letter of commitment (if applicable)